FORMALIZING ANGULAR STRUCTURE OF PARTICLE JETS

By

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and approved by

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ABSTRACT OF THE DISSERTATION

Formalizing angular structure of particle jets

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Prof. Sevil Salur

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Acknowledgments

A nice thank you...

Dedication

A dedication here.

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Chapter 1

Introduction

1.1 The Standard Model

According to the Standard Model (SM), all matter results from interactions between fundamental particles: lepton and quarks (fermions), and mediators (bosons). Table 1.1 shows fermions of the standard model, their anti-particle equivalent would have all the signs reversed for the quantum numbers. For example, an anti-electron (e^+) and anti-electron-neutrino ($\bar{\nu_e}$) lepton number would flip to $(-1, 0, 0)$ and charges (Qs) would be $+1$ and 0 respectively.

	Leptons			Quarks		
	l	Q	(L_e, L_μ, L_τ)	q	Q	$(F_d, F_u, F_s, F_c, F_b, F_t)$
First generation	e^-	-1	(1, 0, 0)	d	$-1/3$	$(-1, 0, 0, 0, 0, 0)$
	ν_e	0	(1, 0, 0)	u	$2/3$	$(0, 1, 0, 0, 0, 0)$
Second generation	μ^-	-1	(0, 1, 0)	s	$-1/3$	$(0, 0, -1, 0, 0, 0)$
	ν_μ	0	(0, 1, 0)	c	$2/3$	$(0, 0, 0, 1, 0, 0)$
Third generation	τ^-	-1	(0, 0, 1)	b	$-1/3$	$(0, 0, 0, 0, -1, 0)$
	ν_τ	0	(0, 0, 1)	t	$2/3$	$(0, 0, 0, 0, 0, 1)$

Table 1.1: Three generations of leptons (left) and quarks (right).

The mediators/bosons exchanged between the other fundamental particles give rise to the fundamental interactions: electromagnetism from the photon (γ), weak force from $W^\pm$ and Z bosons, the gluon (g) for the strong force, and the Higgs boson (h) for imparting mass to the fundamental particles. Various interaction vertices possible in SM, that become the building blocks of all fundamental interactions are shown in 1.2.

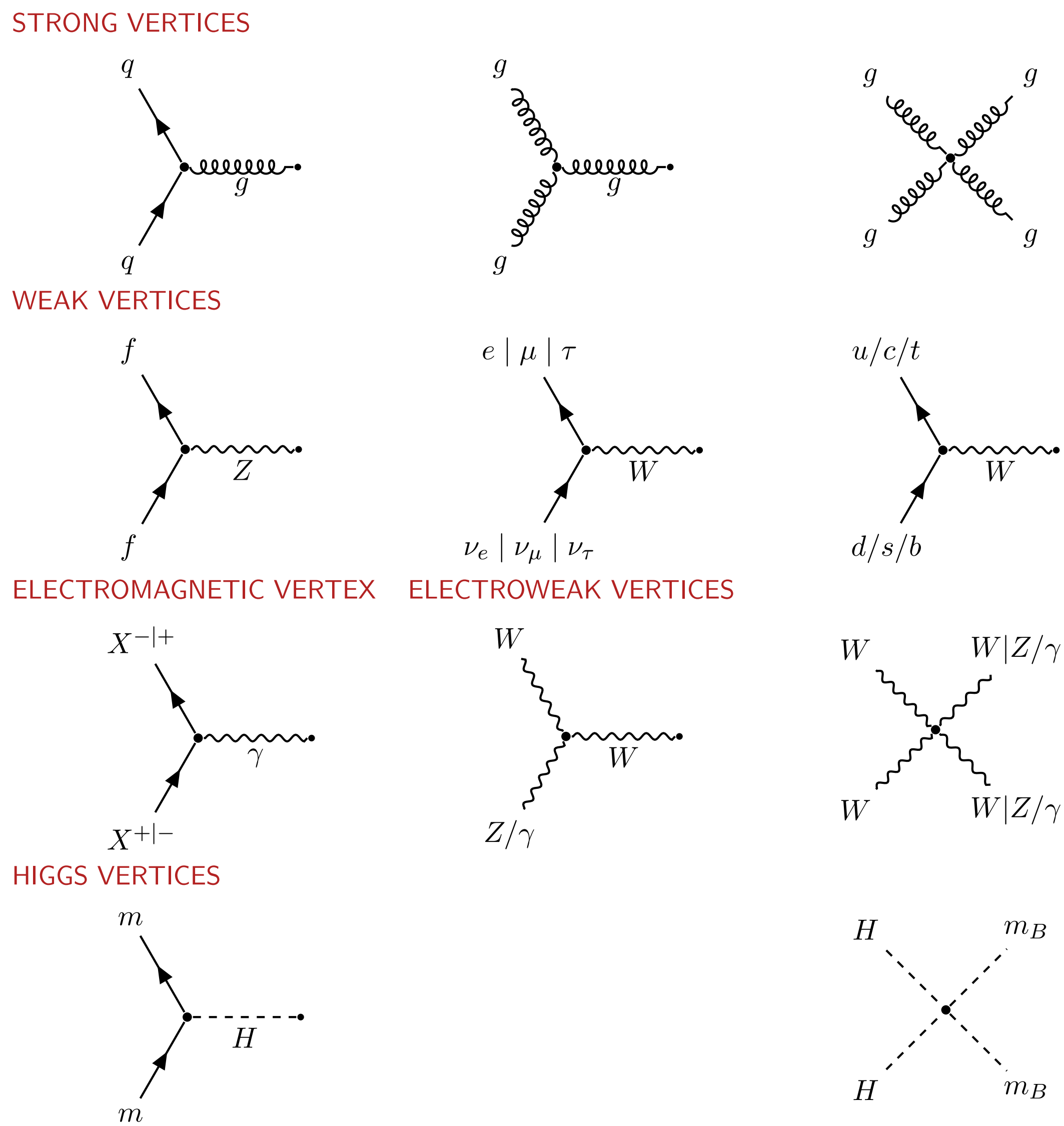

Table 1.2: Feynman diagrams of various interaction vertices possible in the Standard Model

1.2 Quantum Chromodynamics

1.3 Quark-Gluon Plasma

Chapter 2

Relativistic collisions of nuclei

2.1 Relativistic Kinematics

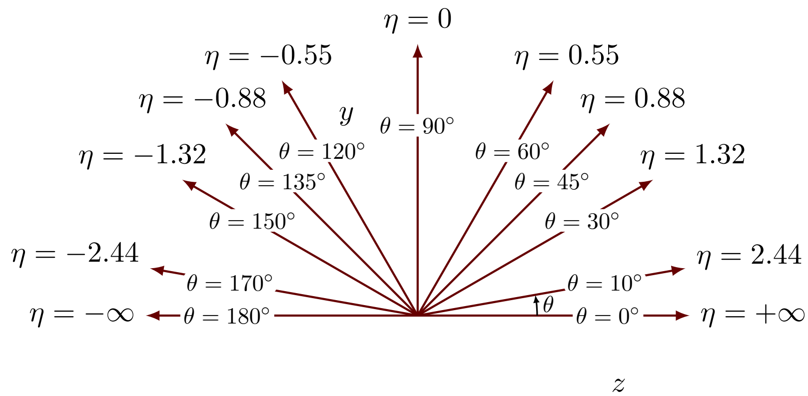

Figure 2.1: η vs θ

2.2 Jets

2.3 Heavy-Ion Collisions

2.4 Signatures of QGP

Different seminal results go here...

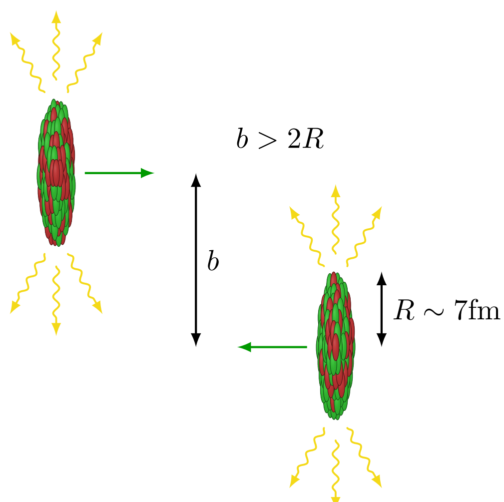

Figure 2.2: Typical heavy-ion collision

Chapter 3

Methodology

Chapter 4

Conclusions & Future Work

I had a really great time in Chapters **??** & **??** & **??**, but I'm quite wearyyyyy.

Bibliography